

Portable Guitar Amplifier with Programmable Effects

James Shanley
Jack Hannon

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Abstract - Modern guitar amplifiers often face tradeoffs between audio fidelity, real-time user control, and portability. This work presents a mixed-signal guitar amplifier which balances these characteristics using analog tone-shaping circuitry, on-board tuning controls, low-latency digital signal processing (DSP) algorithms, and an on-board battery to support standalone operation. The resulting amplifier features a signal-to-noise (SNR) ratio of 65 dB, a maximum average output power of 18.38 W, and a latency of 2.708 ms. Furthermore, low-level control is facilitated through integrated programming ports, which allow users to flash custom audio effects for real-time audio processing. By balancing core performance tradeoffs and extending low-level control, our amplifier serves as a versatile musical tool and an effective educational platform to inspire interest in DSP through the lens of audio.

I. Introduction

The development of high-performance microcontrollers has enabled new approaches to guitar amplifier design. In particular, this shift has supported the development of low-cost, feature-dense systems, which are difficult to realize in traditional analog and vacuum tube amplifiers. By reducing costs, digital processors have expanded the accessibility of audio equipment.

Despite the widespread adoption of DSP in guitar amplifiers, this approach is not without faults relative to traditional analog methodologies. One such advantage of analog circuits is their distinct tonal character, often arising from the nonlinear current-voltage relationships of circuit elements like transistors and operational transconductance amplifiers [1]. The subtle tone shaping and saturation characteristics offered by analog circuits have become coveted by musicians. In response, researchers have developed physics-informed DSP algorithms and mixed-signal circuits to capture the benefits of analog and digital processing techniques for audio applications [1,2,3].

Beyond tonal characteristics, digital amplifiers frequently follow limiting design practices which restrict their versatility. While digital amplifiers have supported the inclusion of complex audio effects, like reverberation and echo, these effects are often provided with fixed parameters and limited user-control. The lack of control over audio effect parameters limits their versatility in musical applications. Similarly, the concept of programmable amplifiers has not been fully explored, despite the inherent programmability of embedded processors.

Several previous works have focused on designing mixed-signal audio systems to address these limitations. [4], [5], and [6] present mixed-signal guitar amplifiers supported by real-time DSP effects. Similarly, [7] and [8] developed multi-effect guitar pedals with rotary encoders for physical control over the audio effect parameters. While these works successfully integrated analog and digital subsystems and provided tunable DSP effects, they came with several limitations related to programmability, portability, and real-time control. For instance, [4] and [6] both used microcontrollers with insufficient memory and processing speed for audio applications. Considering portability, on-board battery units are a valuable addition to audio amplifiers. While the physical footprints of the amplifiers developed in [5], [6], [7], and [8] were compact, these systems lacked on-board batteries, which prevents standalone use. On the topic of real-time control, [4] and [5] relied on a Bluetooth connection to tune the internal parameters of the provided DSP effects. Despite the efficacy of this approach, Bluetooth applications are not favored by

musicians. Rather, musicians tend to prefer simple, physical controls that avoid the complexity and temperamental nature of Bluetooth applications.

This work aims to address the limitations and tradeoffs of modern, commercial, and experimental amplifiers by balancing portability, physical real-time tuning mechanisms, and programmability. The following features and design practices were employed to attain these objectives:

- **Audio Fidelity & Tonal Quality:** A mixed-signal design approach was adopted to balance the tonal characteristics of analog circuits with the versatility of digital processors.
- **Portability:** An on-board battery was included to expand usage settings to outdoor environments.
- **Real-time control:** Physical, analog potentiometers were utilized to provide the user with easy, real-time control over analog and digital audio effects.
- **Programmability:** A widely accessible, user-friendly microcontroller module was selected for digital effects, and user-accessible programming ports were integrated to encourage users to develop custom audio effects.

The overview of our guitar amplifier is outlined in Section II. The design of the analog input circuits is presented in Section III, while the digital audio effects are described in Section IV. The output circuitry is presented in Section V. Hardware design is discussed in Section VI. The results of our experiments are discussed in Section VII, and a conclusion is provided in Section VIII.

II. System Overview and Requirements

Before the design process, system requirements were established to guide the analog circuit design and DSP development. To address portability and user safety, a 12 V DC source was chosen to power the amplifier. This target was chosen to mitigate electrical hazards and align with commercially available batteries. An external DC power supply was chosen to support the on-board battery for extended use cases. To maximize the potential power of the unipolar 12 V source with an Op Amp-based amplifier scheme, our team opted for a 4 Ω speaker to support an average power of 4.5 W (Equation 1).

$$(1) \quad P_{avg} = \frac{V_{rms}^2}{R} = \frac{\left(\frac{12 \text{ V} \div 2}{\sqrt{2}}\right)^2}{4 \Omega} = 4.5 \text{ W}$$

Considering the frequency spectrum of standard 6-string guitars, the frequency range from 80 Hz to 8 kHz was adopted as the focus range to guide the development of filtering stages [9]. Lastly, a target weight limit of 12 lbs was chosen to ensure portability.

The amplifier is defined by the integration of analog and digital subsystems. The full system architecture is outlined in Figure 1. At the start of the signal chain, a monophonic (mono) electric guitar signal is fed in. The signal is immediately buffered, promoting signal integrity and minimizing tonal and amplitude losses. At the output of the buffer, the signal enters a *Gain* potentiometer that modifies the amplitude of the input signal prior to analog waveshaping. After the buffer stage, the signal enters the *Tone Selector* which selects the active *preamp*. The purpose of each preamp circuit is to provide a unique tonal characteristic by altering the audio distortion methodology and frequency content. The preamp stages are termed *Clean*, *Crunch*, and *Fuzz*, in alignment with standard guitar amplifier tone control systems. The *Clean* preamp does not produce

any distortion or clipping, while the *Crunch* and *Fuzz* preamps feature unique waveshaping circuits which add harmonic frequencies to the processed signal.

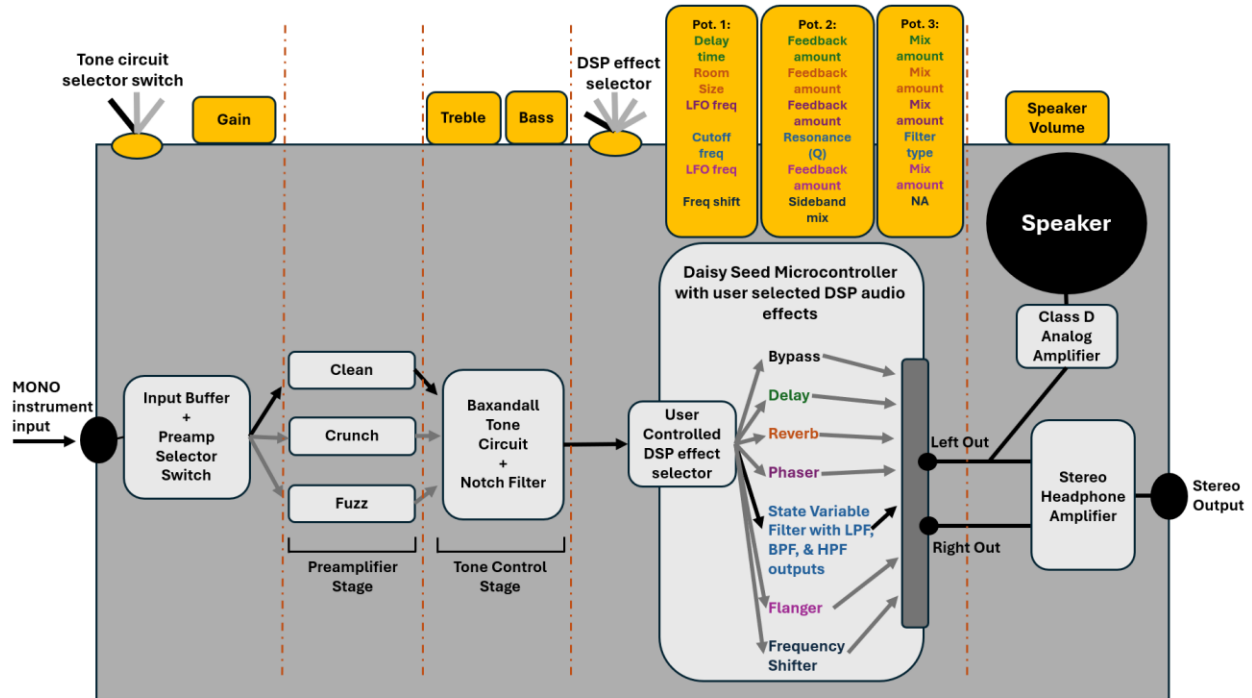


Figure 1: Mixed-signal guitar amplifier schematic.

Immediately following the *Tone Selector/Preamplifier* stage, the signal enters the *Tone Control* stage, which provides independent control over treble and bass frequency content, commonly referred to as “highs” and “lows” respectively.

Ground hum is a common issue for audio equipment which manifests as an audible 60 Hz tone. This arises when using multiple pieces of audio equipment that do not all share the same ground path [10]. To mitigate these effects, a 60 Hz notch filter was adopted in the *Tone Control* stage to selectively attenuate the ground hum signal.

After the analog input sequence, the signal enters a DSP stage, which supports a library of switchable DSP effects (outlined in section IV). To address real-time tunability, analog potentiometers were chosen to control the parameters of the selected DSP effect. Furthermore, our team aimed to design the DSP section to generate a stereo output from the mono input signal to improve sound quality with stereo headsets.

After the DSP stage, the left and right outputs are sent to a stereo headphone amplifier that supports a line level output signal suitable for low-impedance headphones and high-impedance audio interfaces. The mono speaker signal was taken from the left stereo channel, and a *Speaker Volume* control was chosen to alter the volume of the speaker stage. Additionally, a speaker switch control was chosen to turn the speaker output on and off.

III. Analog Preamplifier Design

A. Power

The first consideration of the amplifier was power management. A commercial 12V Lithium-ion battery pack with 3000 mAh of capacity and an integrated battery monitoring system (BMS) was chosen to power the amplifier. The integrated BMS has overcurrent protection for maintaining battery health in extenuating circumstances [11].

Figure A.1 (Appendix) displays the power management circuitry. A 5 Ampere fuse was included to protect the internal circuitry from overcurrent, and mechanical switching logic was implemented to prioritize the use of the external power supply when connected (Appendix, Figure A.1 left). Additionally, a transient-voltage-suppression diode was included to protect against electrostatic discharge and reverse-polarity power. To support analog waveshaping of the AC audio signal with the single-sided supply, a steady 6 V virtual ground was created and buffered for stability (Appendix, Figure A.1 center).

B. Input buffer

Electromagnetic guitar pickups have an inherent RLC impedance [12]. When a guitar pickup is connected to a circuit with input impedance Z_{in} , the output voltage is

$$(2) \quad V_{out} = V_{pickup} \left(\frac{Z_{in}}{Z_{in} + Z_{pickup}} \right),$$

considering the guitar's impedance as Z_{pickup} and the voltage generated by the pickup as V_{pickup} . Smaller input impedances result in smaller output amplitude. Since guitar pickups are highly inductive, this effect worsens at higher frequencies, resulting in undesirable tonal changes [12]. The Input Buffer stage (Appendix, Figure A.2) prevents this issue by employing a high-impedance, JFET Op Amp (TL072) input stage with a low output impedance [13]. Additionally, The Class AB amplifier output of the TL072 minimizes switching noise. These beneficial characteristics made the TL072 the standard Op Amp throughout our amplifier.

Since guitar pickups are electromagnetic, they are susceptible to radio-frequency interference. To mitigate this noise, a 1 nF capacitor was placed at the input to form a lowpass filter with the guitar pickups' internal impedance [14]. Considering that the pickup impedance of electric guitars typically ranges from 5 k Ω to 20 k Ω , the resulting lowpass filter exhibits a cutoff range from approximately 31.8 kHz to 7.96 kHz, which is sufficient to attenuate radio frequency interference without significantly affecting the guitar signal (80 Hz to 8 kHz). A series 100 Ω resistor was used to prevent overloading of the Op Amp and oscillations due to parasitics. Furthermore, a gain potentiometer at the output allows the user to vary the gain of the guitar signal from zero to one, which plays an important role in altering the preamp sound. After this gain stage, the signal enters the tone selector switch, where it is routed to one of three preamps circuits: *Clean*, *Crunch*, or *Fuzz*.

C. Preamplifier - 'Clean' and 'Crunch' Circuits

The *Clean* and *Crunch* preamp circuits (Figure 2) both utilize the TL072 in a non-inverting configuration. The *Clean* preamp boosts the signal with a DC gain of 11, which was qualitatively determined to provide a clean, undistorted tone. The *Crunch* preamp is of the same topology,

though the gain is increased to 86 to cause intentional clipping at the 12 V rail. This gain value was selected through qualitative testing, where a gain factor of 86 yielded distorted tonal qualities similar to commercial amplifiers. In each preamp, small Miller feedback capacitors were added to prevent high-frequency oscillations [15].

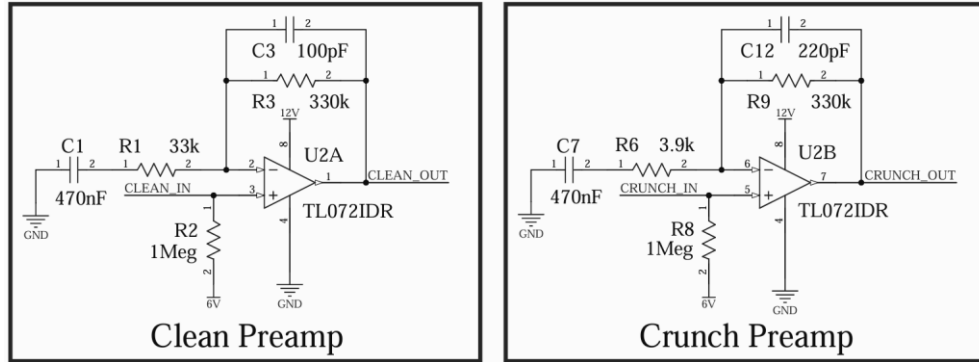


Figure 2: ‘Clean’ and ‘Crunch’ preamplifier circuits.

D. Preamplifier - ‘Fuzz’ Waveshaping Circuit

The custom *Fuzz* preamp topology greatly differs from the *Clean* and *Crunch* circuits in that it uses cascading transistor amplifiers to amplify and saturate the incoming signal (Figure 3). By employing transistor-based saturation rather than diode clipping, the *Fuzz* circuit introduces harmonic content unique from the *Clean* and *Crunch* preamps. In the circuit, 4.7 μ F capacitors were included at each emitter node to allow audio-range frequencies to bypass the emitter resistor and thus expand the gain across a wide frequency range. During testing, C14, C18, and R20 were added to maintain stability and mitigate high-frequency self-oscillation.

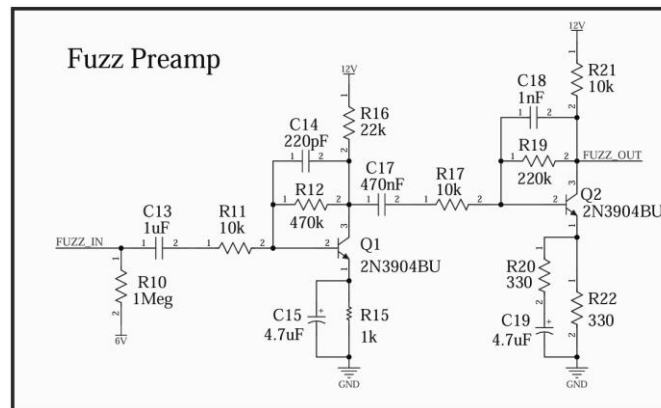


Figure 3: ‘Fuzz’ preamplifier circuit.

E. Baxandall and Notch Filter

After the preamp stage, the signal is routed through a buffer and a diode clipping network (Figure 4 left). While the buffer maintains the preamp signal’s waveform and removes any interaction between the preamps and future stages, the diode clipping network shapes the audio waveform to introduce new harmonics and enforce amplitude bounds from -1.4 V to +1.4 V, which

are necessary for the safety of the microcontroller in the DSP stage. Due to the gain differences in the preamp stages, the *Clean* preamp signal is left undistorted for moderate input gain values, while the *Crunch* and *Fuzz* signals undergo varying levels of distortion.

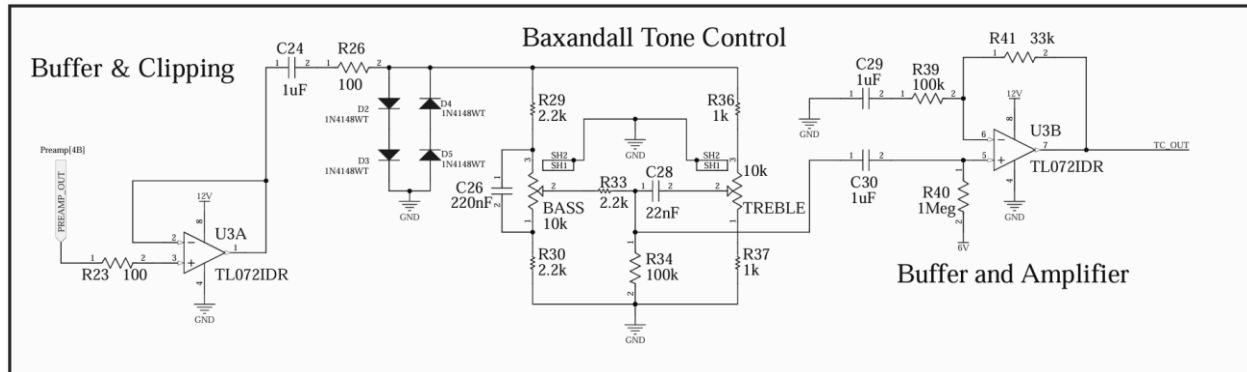


Figure 4: Tone control circuit.

Next, the signal enters a passive *Baxandall Tone Control* (Figure 4). This circuit uses two potentiometers for independent control of treble and bass frequencies [16]. The responses can be described as “shelves,” wherein all frequencies above or below the center frequency are shifted up or down in magnitude, rather than tuning a cutoff frequency value. For an electric guitar, the majority of the frequency content is centered around the “mid” frequencies (250 Hz to 4 kHz) [14]. Considering this range, our team tuned the Baxandall center frequency to 650 Hz. The resulting filter offered distinct signal clarity compared to other amplifiers with different center frequencies. Figure 5 shows the full versatile range of both the treble and bass controls.

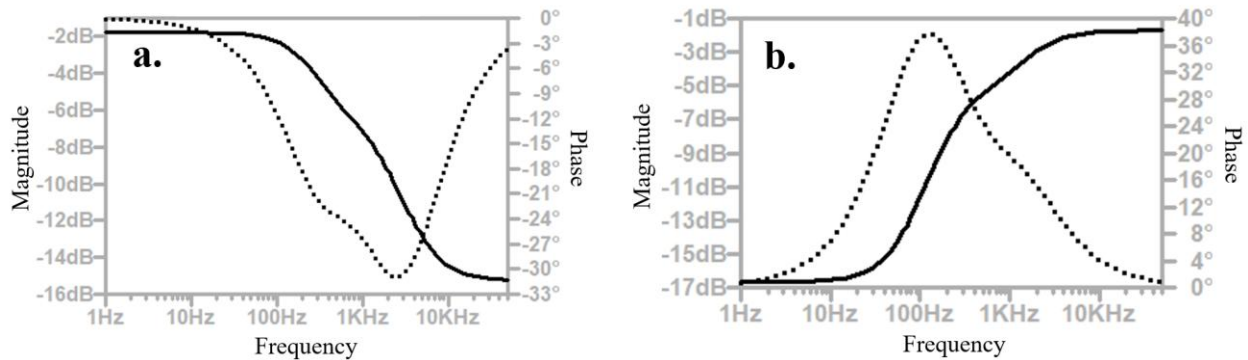


Figure 5: Frequency response of Baxandall tone circuit at various settings. (a) maximum bass and minimum treble levels. (b) minimum bass and maximum treble levels. (solid line = magnitude response, dotted line = phase response).

Due to the passive nature of the Baxandall circuit, there is some signal loss. With maximized bass and treble levels, the signal exhibits a maximum magnitude of approximately -1.8 dB. The signal loss was accounted for by introducing a gain of 1.33 in the Baxandall output buffer (Figure 4 right). The final analog circuit before the DSP stage is a 60 Hz *Notch Filter* for eliminating ground hum (Figure 6) [4]. Precise tuning of the cutoff frequency was supported by trimmer potentiometers, which may be adjusted over the range from 50 Hz to 60 Hz to target ground noise.

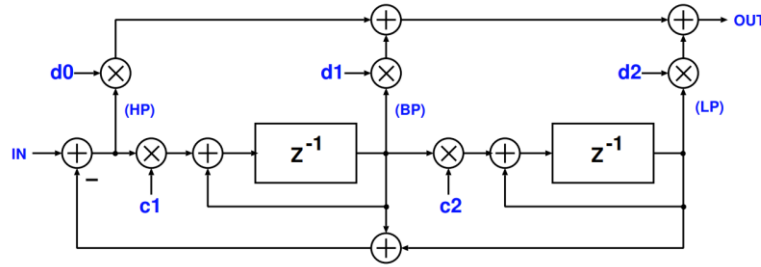


Figure 7: Digital state variable filter diagram [19].

The state variable filter proposed in [19] (Figure 7) was expanded for real-time use on the DS microcontroller. Real-time control was achieved by mapping potentiometer voltages to the cutoff frequency, resonance/quality factor (Q), and the active filter output. A tunable cutoff frequency range from 80 Hz to 8 kHz was selected to reflect the frequency range of electric guitar signals. The resonance potentiometer was configured to support a variable gain of up to 10 dB at the corner frequency. The active filter output was determined by establishing three unique potentiometer voltage regions corresponding to LP, BP, and HP outputs. This was accomplished by switching the Biquad coefficients derived in [19]. Furthermore, all potentiometer values were smoothed with a rolling average filter to eliminate noise. Smoothing filters were included in all DSP effects to stabilize potentiometer values.

B. Delay

Delay and echo effects are among the most commonly used effects for guitars. The basis for our delay effect was inspired by the fractional delay line structure in [20]. Fractional delay lines expand tunability by supporting non-integer delay lengths. Our delay effect builds off of this structure to provide control over delay time, feedback amount, and mix amount, which are all critical parameters for musical delays.

When the delay effect is active, the amplifier's DSP tuning potentiometers are linked to the delay length, feedback amount, and mix amount. The delay length potentiometer controls the offset between the read and write head of a circular buffer, thereby controlling the period of the echo from 0.2 ms to 4 seconds. This control was mapped to the delay time according to a parabolic relationship to offer fine-tuning at small delay times and coarse tuning at longer delays. The feedback potentiometer was used to control the number of echo repeats, while the mix control was used to tune the relative amplitude of the input and echo signal.

Several other structural features were included to improve the fractional-delay design. Hermite interpolation was employed to determine the amplitude of the delay line output at fractional indices. This technique offers improvements in signal estimation over linear interpolation at a small computational cost [21]. Furthermore, a lowpass filter and saturation function were added to the feedback path to attenuate high frequency content in the echo signal and mitigate runaway self-oscillation at high feedback gains (Figure 8). In the final implementation, two delay objects were performed in parallel to support a stereo output.

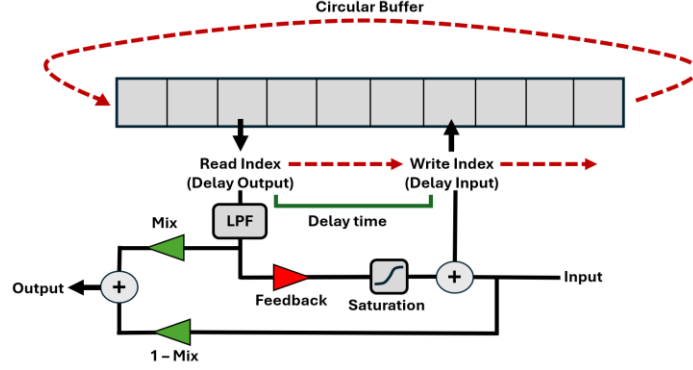


Figure 8: Fractional delay (echo) diagram.

C. Reverberation

Reverberation algorithms are used by musicians to mimic the effect of sounds propagating in physical spaces, whether it be a large cathedral or a small room. Research into reverberation algorithms has focused on expanding tunability, obtaining long decay envelopes, and avoiding resonant frequencies arising from the feedback of short delay lines [22, 23]. While many efficient algorithms have been designed [22], the feedback delay network (FDN) proposed in [24] offers a beneficial foundation to achieve these goals. This structure uses a parallel series of delay lines which are mixed using a Hadamard matrix before feedback to the delay line inputs [24]. Mixing the delay signals before feedback aids in reducing resonant frequencies. Furthermore, this structure maintains stability at unity feedback and offers control over the reverb decay length using a feedback gain parameter [22].

The reverberation algorithm developed for our guitar amplifier was inspired by the modified FDN structure presented in [22] and [25]. In particular, these papers inspired the integration of All-Pass Filters (APFs) and a global delay time parameter. In this context, APFs serve as short, frequency dependent delays which dampen resonant frequencies [26]. Since the delay lengths are analogous to the dimensions of a physical space, the inclusion of a delay time factor allows the user to change the ‘size’ of the artificial room for different musical applications.

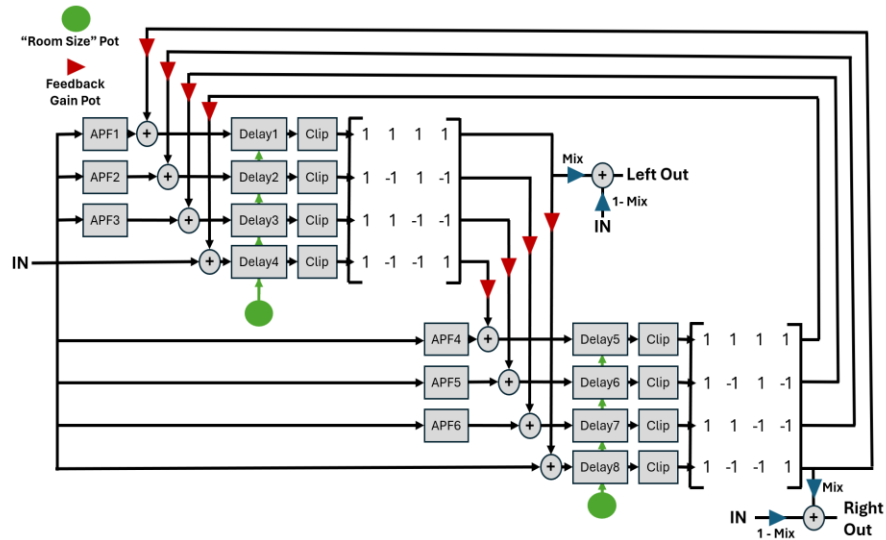


Figure 9: Reverberation algorithm diagram.

Our reverb algorithm expands the traditional FDN structure by cascading two FDN stages and integrating modern features like APFs and a global delay time control (Figure 9). In this context, the global delay parameter is used to scale the delay lengths by a constant factor. This method maintains the target, mutually prime relationships between each delay length, a necessary step to mitigate resonant frequencies. Similar to previous effects, mix and feedback potentiometers were included. Unique stereo outputs were taken from the output of each FDN stage.

D. Frequency shifter & Ring Modulator

The frequency shifter uses single-sideband modulation techniques to increase or decrease all frequencies of the input signal by a uniform amount. This is achieved by passing the input ($m(t)$) through a Hilbert transform to generate a quadrature signal ($m_h(t)$), before multiplying the pair by a sinusoidal quadrature oscillator and calculating the sum and difference of the products to isolate the lower- and upper-sidebands (Equation 2) [27]. Ring modulation can be obtained by summing the upper- (USB) and lower-sidebands (LSB) with equal weights [27].

$$(3) \quad \begin{aligned} USB(t) &= m(t)\cos(2\pi f_c t) - m_h(t)\sin(2\pi f_c t) \\ LSB(t) &= m(t)\cos(2\pi f_c t) + m_h(t)\sin(2\pi f_c t) \end{aligned}$$

Our frequency shifter algorithm utilizes a 501-tap Hilbert transform FIR filter to generate a phase shifted copy of the input signal, and a 251-tap circular buffer to equate the group delays of the input and Hilbert transform signals (Figure 10). We used a through-zero quadrature oscillator to achieve positive and negative frequency shifting of each sideband. The algorithm was tuned using potentiometers to control the frequency shift amount and the sideband mix.

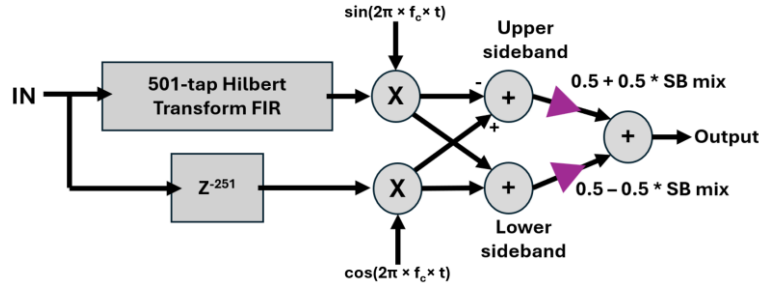


Figure 10: Hybrid frequency shifter/ring modulator diagram.

A cubic function was used to map the frequency potentiometer value to the quadrature oscillator frequency. This mapping supports fine tuning at low oscillator frequencies and coarse tuning at high frequencies (-500 Hz to +500 Hz). The sideband mix potentiometer controls the relative weight of the USB and LSB during the final summation. Since the quadrature oscillator supports positive and negative shifting of each sideband, the potentiometer was configured to indicate equal sideband weighting (ring modulation) at its minimum position and isolation of the USB at its maximum position.

V. Power Output Design

The final stage of the guitar amplifier includes the headphone and speaker outputs. The speaker amplifier is based on the TPA3112D1 Class-D switching amplifier, which features a high

input impedance and a modulation scheme that supports output voltages from -12 V to +12 V with our power supply (Appendix, Figure A.3-4) [28]. By doubling the output voltage range, the Class-D amplifier allowed our amplifier to achieve a maximum theoretical average power four times greater than our initial target of 4.5 W (Equation 4).

$$(4) \quad P_{avg} = \frac{V_{rms}^2}{R} = \frac{\left(\frac{12 V}{\sqrt{2}}\right)^2}{4 \Omega} = 18 W$$

The amplifier IC was limited to 20 W using a stable reference voltage to prevent speaker damage during faults. In addition, an RC snubber along with a ferrite bead and capacitor network was placed on both sides of the amplifier output to remove RF noise and combat parasitics [29].

To complement the speaker amplifier, a line-level headphone amplifier was designed to support the DSP's stereo output signal (Appendix, Figure A.5). This circuit uses OPA1688 Op Amps, which are capable of supplying up to 75 mA for driving headphones and other lower-impedance loads [29]. Large 220 μ F capacitors isolate the DC level of the amplifier from the headphones without audibly impacting the frequency content of the guitar signal.

VI. Hardware Design

After validating the analog and digital subsystems in isolation, a printed circuit board (PCB) was designed to optimize signal integrity and longevity of the final product (Figure 11). The PCB features a 4-layer stackup: (bottom) signal, ground plane, power routes, and signal (top). During the design process, careful consideration was given to PCB layout techniques to mitigate electromagnetic interference (EMI) and improve signal integrity. Specifically, the board was laid out to minimize trace length and optimize heat dissipation. Furthermore, a ground pour was created on the top layer to reduce EMI.

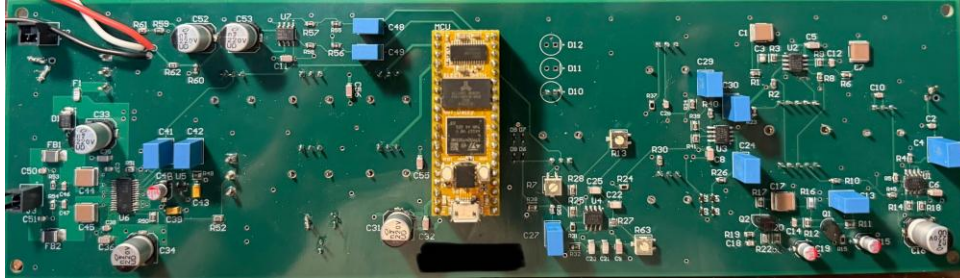


Figure 11: Guitar amplifier PCB.

The enclosure for our amplifier was designed considering portability, user-accessible controls, and programmability (Figure 12). All potentiometers and switches were mounted on the top panel to provide the user with easy access to key tuning parameters. On the back panel, audio jacks were mounted for the input and output signal, in addition to ports for an external DC power adapter and battery charger. To support programmability without disassembly, our team mounted a USB port connected to a microcontroller via an ST-LINK programmer on the back panel.



Figure 12: Guitar amplifier enclosure.

VII. Results and Discussion

To validate the analog circuits, each preamp was simulated along with all circuitry preceding it, with the simulated waveforms then compared to measured waveforms on the final PCB. The final PCB was tested using a steady 12 V power supply. A 500 Hz sinusoid with a 500 mV amplitude was input to the circuit, and the results are shown below (Figures 13).

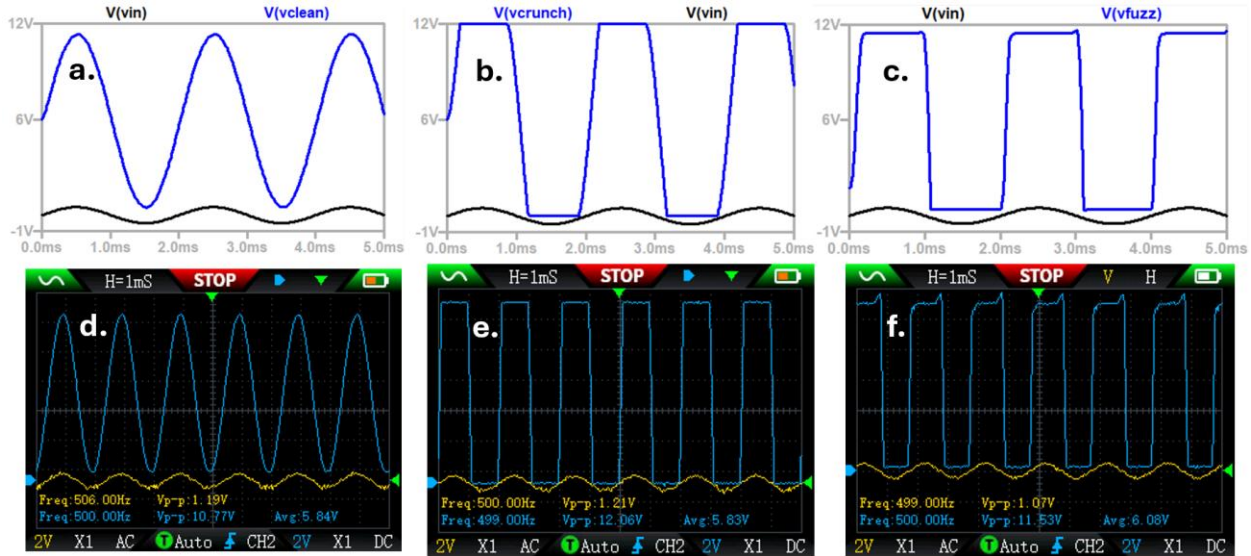


Figure 13: (a-c) Simulation of Clean, Crunch, and Fuzz preamps, respectively. (d-f) Experimental waveforms of Clean, Crunch, and fuzz preamps, respectively.

Each preamp circuit produced an output waveform in close alignment with the simulated signal. The clean preamp circuit provided an amplified output signal with no clipping (Figure 13.a,d). In contrast, the Crunch and Fuzz preamp circuits both distorted the sinusoidal input signal to produce the target waveshaping characteristics (Figure 13). Aside from a small difference in the signals' headroom, the simulations and measurements were nearly identical.

The cutoff frequency range and magnitude response of the state variable filter were assessed to confirm its functionality. White noise was input into the Daisy Seed's input channel. The processed audio signal was passed to MATLAB using an Arturia Minifuse audio interface, where FFT was performed to generate a plot of the signal power (dBm) versus frequency (kHz) (Figure 14). The lowpass, highpass, and bandpass were confirmed to share a common cutoff frequency. Additionally, no attenuation was observed in the passband of each filter output.

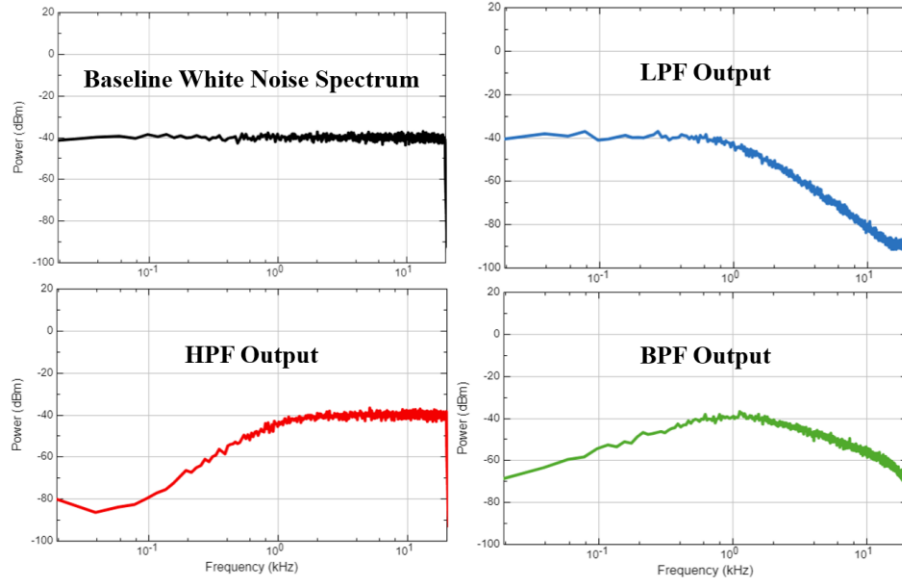


Figure 14: State variable filter frequency spectra at $f_c = 1$ kHz.

The cutoff frequency range was determined by tracking the peak of the bandpass filter output at the extreme potentiometer positions. The observed cutoff frequency range spans from 78.08 Hz to 8.07 kHz. Considering the target range from 80 Hz to 8 kHz, these experimental values correspond to errors of -2.4% and +0.87%, respectively. To evaluate the parabolic delay time tracking, the normalized potentiometer voltage was compared to the experimental delay time (Figure 15). The experimental delay range was determined to be 0.24 ms to 3.996 s, which closely aligns with the target range from 0.21 ms to 4.0 s.

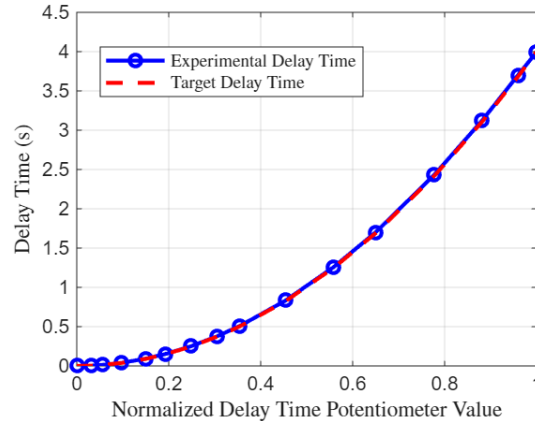


Figure 15: Comparison of target and experimental delay time tracking.

The frequency shifter was verified following a similar procedure. A 1 kHz sine wave signal was input to the microcontroller, and a FFT was performed on the output to quantify the frequency shift and sideband suppression. At the extreme frequency shift values (± 500 Hz), sidebands were observed at 507.5 Hz and 1.503 kHz. Considering the true input frequency (995.5 Hz), the error in the frequency shifter is 1.5% for positive frequency shifts and 2.4% for negative frequency shifts. Since $\Delta \text{dB}_m = \Delta \text{dB}$, the sideband suppression of the frequency shifter algorithm may be determined from the power spectra. The average sideband suppression was calculated to be 49.1 dB.

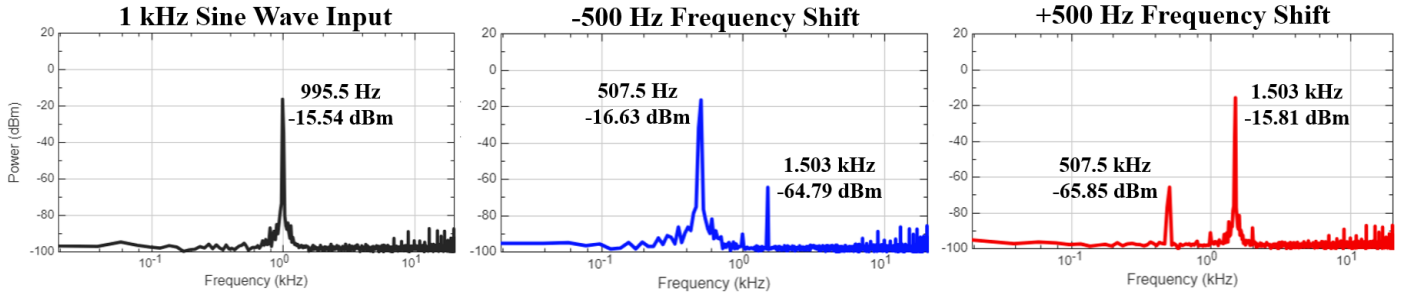


Figure 16: Frequency shifter sideband suppression and tracking results.

After assembling the integrated system, the signal-to-noise ratio was evaluated according to the procedure outlined in [30]. A 1 kHz sine wave was input to the system, and the output was monitored from the stereo output. Using Equation 5, the SNR was determined to be 65 dB.

$$(5) \quad SNR (dB) = 20 \log_{10} \left(\frac{RMS \text{ Signal}}{RMS \text{ Noise}} \right)$$

Furthermore, the complete amplifier supported a maximum average power consumption of 18.38 W, closely aligning with the theoretical output power of 18 W. This corresponds to a peak RMS current draw of 1.532 A. When considering the 3000 mAh battery, the amplifier's minimum theoretical battery runtime is 1.96 hours. Direct comparison to related guitar amplifiers [4-8] is limited by the lack of quantitative performance metrics in these studies. Therefore, to provide some basis for comparison, our team evaluated our guitar amplifier against commercial alternatives (Table 1).

Table 1: Summary of Experimental Results

	Commercial Portable Amplifier Range	Target Metric	Experimental Result
SNR (dB)	~50 - 90 [31]	60	65.0
Output Power (W)	~3 - 25	18	18.38
Latency (ms)	2 - 10 [32]	< 10	2.708
Weight (lbs)	~2 - 12	12	5.4

The experimental results indicate that the proposed guitar amplifier is competitive with commercial alternatives in terms of SNR, output power, latency, and weight. Despite the similarities in performance metrics, our design expands on the functionality of modern amplifiers by supporting low-level control via a user-accessible programming port and open-source DSP effects. Thus far, a total of six effects have been developed for the amplifier, consuming 62.92% of the Daisy Seed's flash memory and 2.92% of its SDRAM. Considering these results, there are substantial memory resources remaining for future users to develop custom algorithms for audio processing.

VIII. Conclusion and Future Work

This work presents a mixed-signal guitar amplifier design with emphasis on audio fidelity, portability, real-time control, and programmability. By strategically integrating analog and digital subsystems, the resulting amplifier balances the tonal characteristics of analog waveshaping circuits and the versatility of high-performance microcontrollers. The resulting system achieved SNR (65 dB), output power (18.38 W), and latency (2.708 ms) performance metrics consistent with commercial amplifiers, while providing nuanced control over the amplitude, frequency content, and timbre of the guitar signal through user-accessible controls. Furthermore, these capabilities were accomplished within a portable form-factor, with overall dimensions of 13" $\times$ 7" $\times$ 8", a total weight of 5.4 lbs, and an on-board rechargeable battery to support standalone operation.

Experimental results demonstrated that the analog and digital subsystems were closely aligned with the target metrics. The preamplifier circuits each produced waveform outputs consistent with simulated results. Considering the digital multimode filter, accurate LP, HP, and BP outputs were observed at a common cutoff frequency, and the cutoff frequency range remained within 2.5% of the target. Similarly, precise delay time tracking and frequency shifting were confirmed for the delay and frequency shifter effects.

Future work will focus on expanding the open-source DSP library and including helpful documentation. Our team aims to encourage musicians and engineering students to learn about digital signal processing through the lens of audio, and we expect that user-friendly documentation will aid in achieving this goal. Additionally, performance metrics including total harmonic distortion and battery life will be quantified in future work.



Figure 17: Completed guitar amplifier in use.

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Appendix

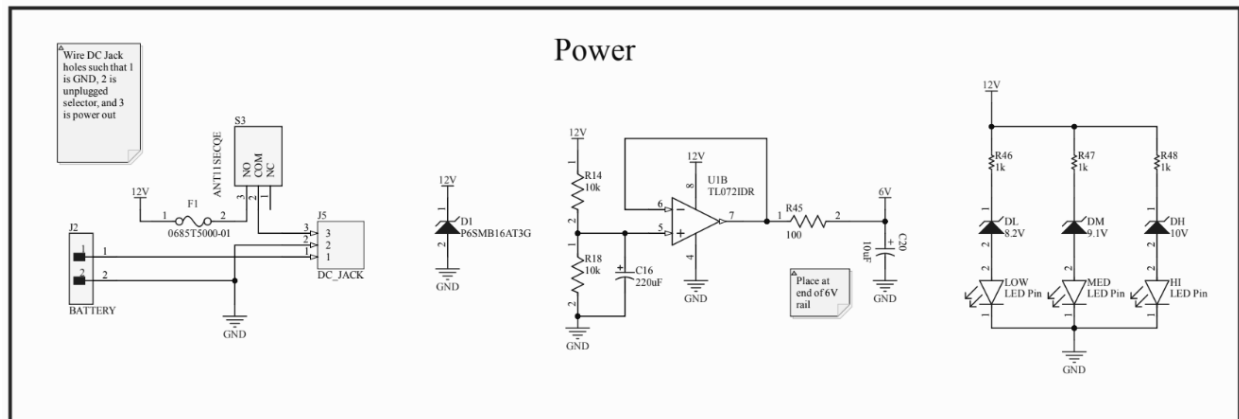


Figure A.1: Analog power circuits

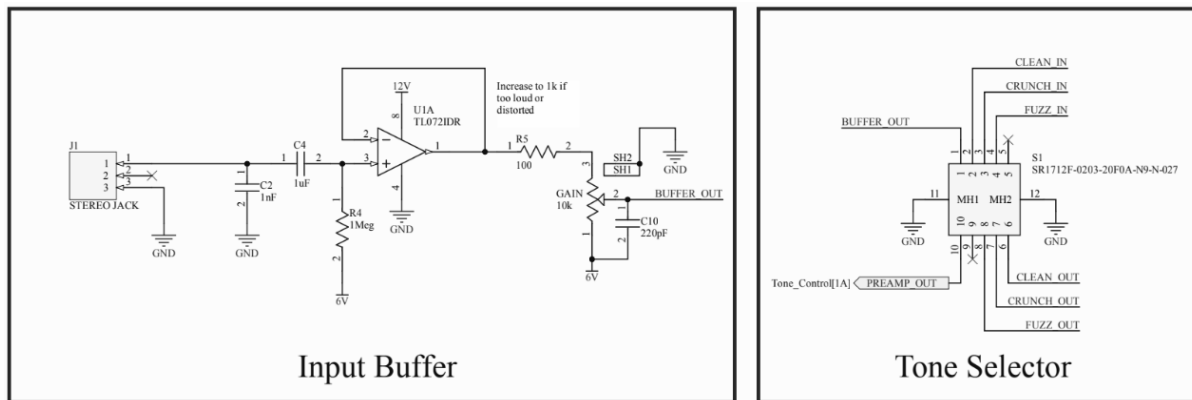


Figure A.2: Analog Input buffer stage (left) and preamplifier tone control circuit selector (right)

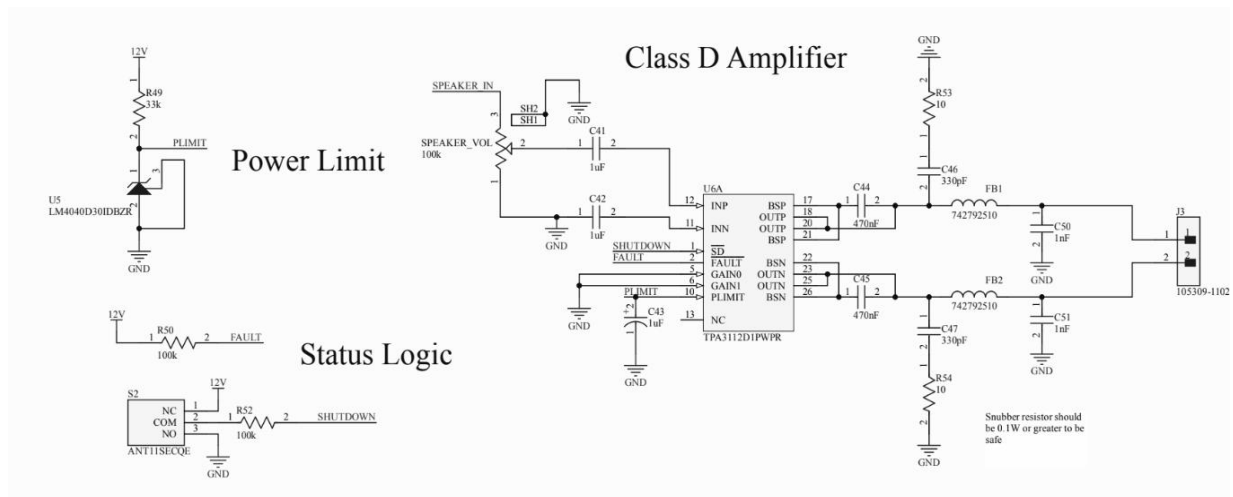


Figure A.3: Class-D output amplifier circuit

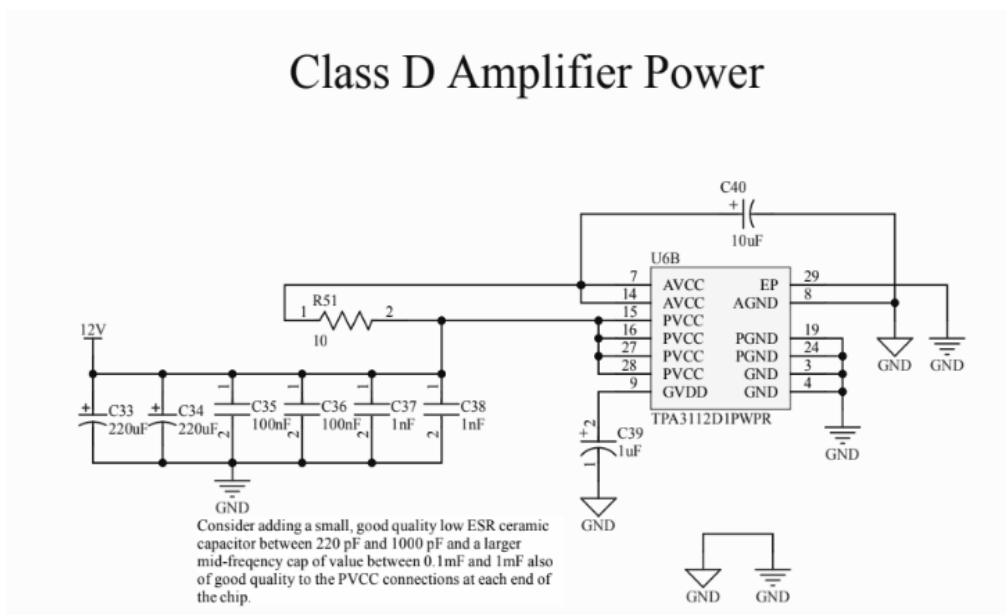


Figure A.4: Class-D output amplifier power circuit

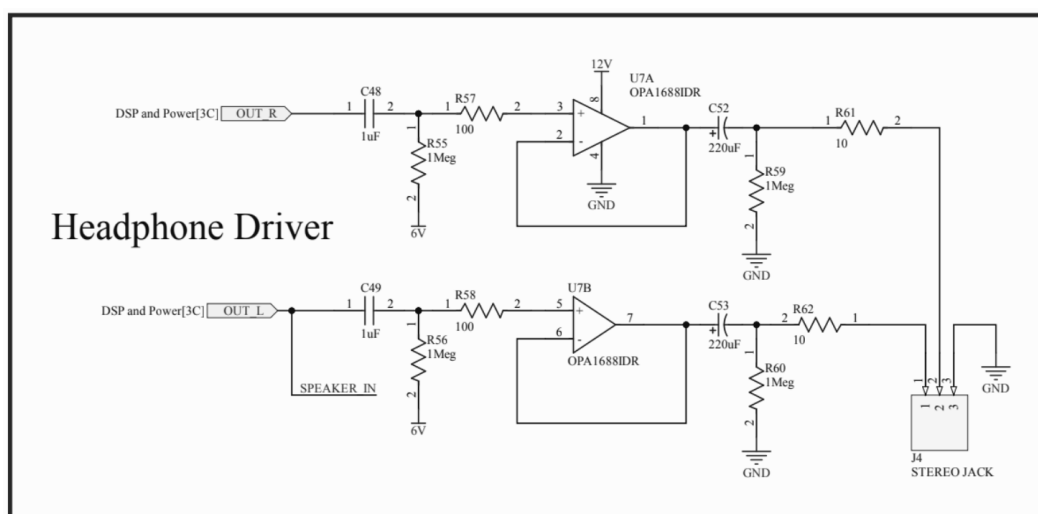


Figure A.5: Stereo headphone driver circuit.